Online Retail Sales Analysis

Dataset Source:

The Online Retail dataset was drawn from UCI Machine Learning Repository. It is described by UCI as transaction data for a non-store online retailer based in the UK, for the period 1 December 2010 to 9 December 2011. The excel file downloaded has 541,909 rows of transactions. The original worksheet was renamed Raw_Data and kept as the unedited source. A copy of Clean_Data was created for all subsequent preparation and analysis

Data Cleaning and Preparation:

When checking for exact duplicate rows, the original eight fields were used. The Remove Duplicates command was applied to the Clean_Data copy and the Raw_Data sheet was left untouched. Missing identifiers and descriptions were left as is, but marked. Unit prices less than or equal to zero were considered invalid. An invalid-price rule was applied and then the invoice number was C or the quantity was negative and it was determined to be a valid cancellation.

Missing CustomerID records were retained in overall sales, product, country and monthly analysis because they are valid transactions. Such purchases could not be reliably attributed to a specific customer and therefore were not included in the customer-level analysis. Named-product rankings did not include missing descriptions, but did include them for reconciliation totals.

Cleaning formula:

=IF([@CustomerID]="","Missing Customer","Known Customer")

=IF([@[Description]]="","Missing Description","Available")

=IF([@[UnitPrice]]<=0,"Invalid Price",IF(OR(LEFT([@[InvoiceNo]]&"",1)="C",[@[Quantity]]<0),"Cancellation","Sale"))

Month_Year as text, it was stored as a real date using the first day of each month. This guarantees that months are sorted chronologically in PivotTables and charts. The cell was formatted using a custom number format of mmm-yyyy.

Monthly Sales Analysis:

I made a PivotTable with Month_Year in Rows and Transaction_Type filtered to Sale. Sales_Revenue and Quantity were summarised with sum. InvoiceNo and CustomerID were added via the Data Model to allow the use of Distinct Count. The average order value was calculated as the monthly sales revenue divided by the number of distinct sale invoices.

Transaction_Typ	Sale			
Row Labels	**Number of Invoices**	**Monthly Revenue**	**Quantity Sold**	**Average Order**
December	**66,591.00**	**£1,462,538.82**	**673487**	£21.96
2010	41,480.00	£823,746.14	359239	£19.86
2011	25,111.00	£638,792.68	314248	£25.44
November	**83,369.00**	**£1,509,496.33**	**754507**	£18.11
2011	83,369.00	£1,509,496.33	754507	£18.11
October	**59,304.00**	**£1,154,979.30**	**623401**	£19.48
2011	59,304.00	£1,154,979.30	623401	£19.48
September	**49,261.00**	**£1,058,590.17**	**570820**	£21.49
2011	49,261.00	£1,058,590.17	570820	£21.49
August	**34,483.00**	**£759,138.38**	**421770**	£22.01
2011	34,483.00	£759,138.38	421770	£22.01
July	**38,645.00**	**£719,221.19**	**401759**	£18.61
2011	38,645.00	£719,221.19	401759	£18.61
June	**35,977.00**	**£761,739.90**	**389213**	£21.17
2011	35,977.00	£761,739.90	389213	£21.17
May	**36,164.00**	**£770,536.02**	**395738**	£21.31
2011	36,164.00	£770,536.02	395738	£21.31
April	**29,096.00**	**£537,808.62**	**308815**	£18.48
2011	29,096.00	£537,808.62	308815	£18.48
March	**35,803.00**	**£717,639.36**	**377526**	£20.04
2011	35,803.00	£717,639.36	377526	£20.04
February	**27,105.00**	**£523,631.89**	**283555**	£19.32
2011	27,105.00	£523,631.89	283555	£19.32
January	**34,306.00**	**£691,364.56**	**387785**	£20.15
2011	34,306.00	£691,364.56	387785	£20.15
Grand Total	**530104**	**10,666,684.54**	**5588376**	
				£507.61

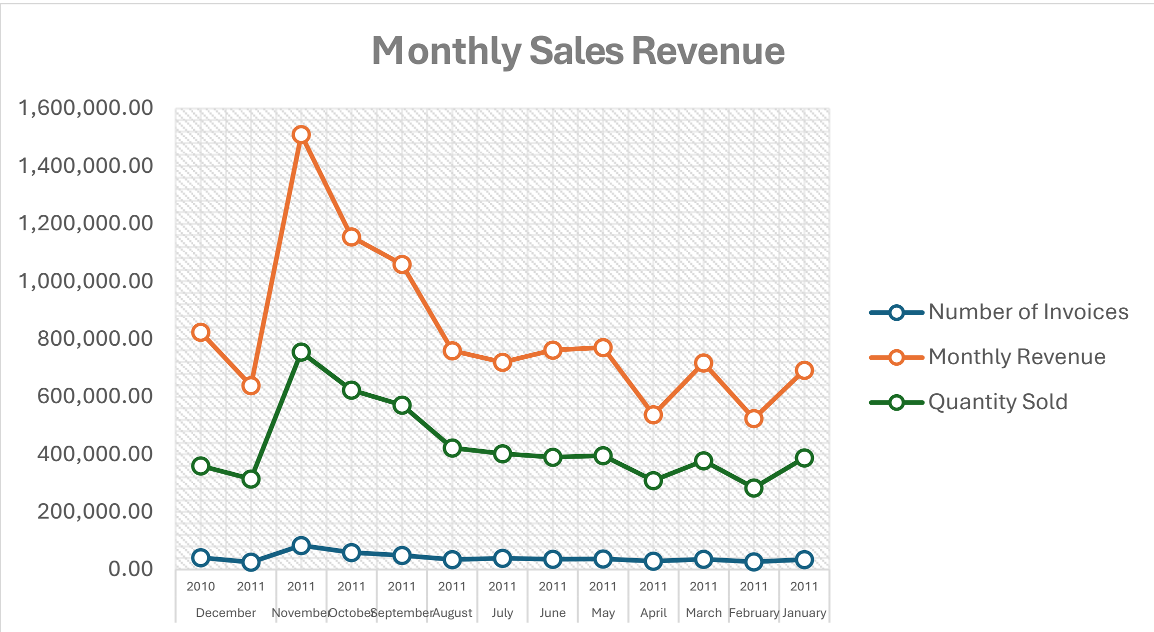


Fig1: Monthly sales revenue and net revenue after duplicate removel.

The highest sales revenue was in November 2011 at £1,503,866.78 and the highest net revenue was in November 2011 at £1,456,145.80 . Revenue increased significantly from

September to November, in line with the normal end-of-year peak. The lowest full month for sales revenue was £522,545.56, February 2011. December 2011 contains only transactions to 9 December and its totals are therefore not comparable to full months. It also had the highest cancellation value, £205,089.27, which dropped its net revenue to £432,701.06. Overall the average order value was £533.17.

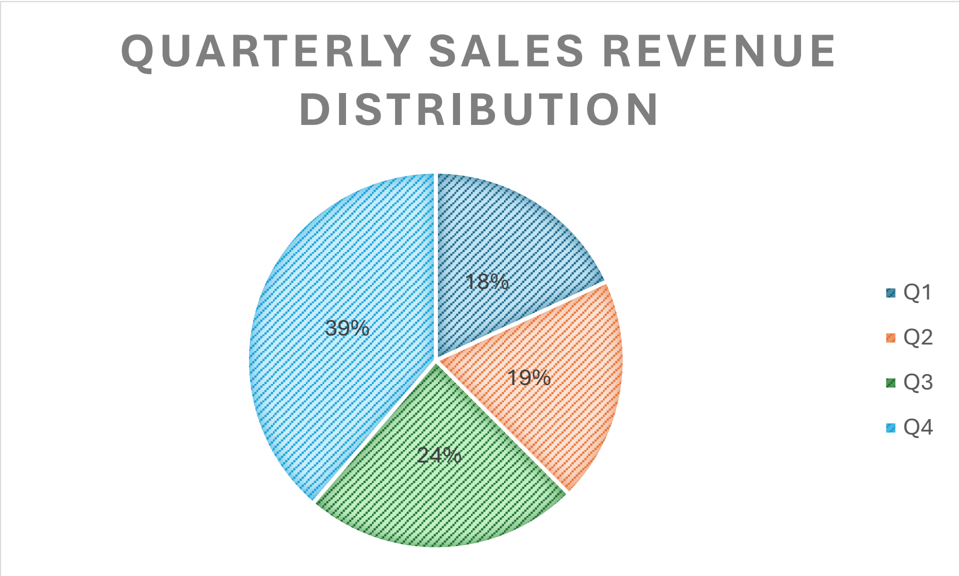
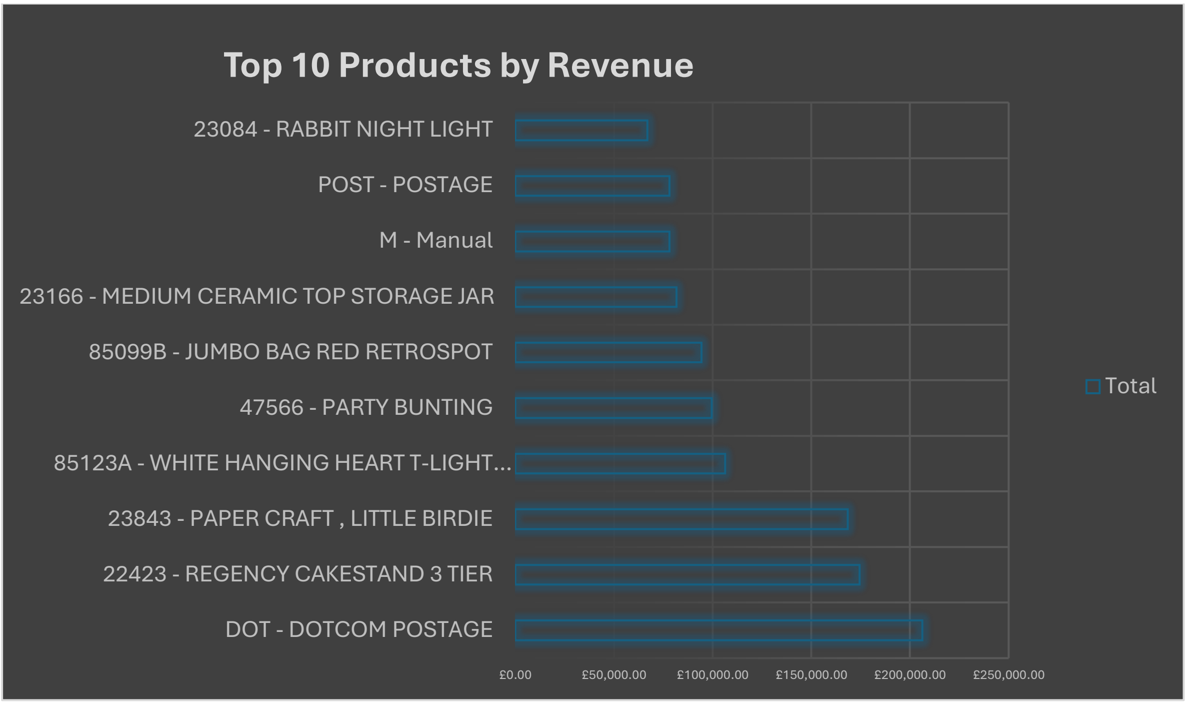


Fig2: Quarterly distribution of sales revenue Q4 for 39% of total sales revenue

Product Analysis

Product_Key was used to assess product performance, which was a combination of StockCode and Description after trimming. Revenue, quantity, cancellation quantity, cancellation value and average unit price were created as separate Pivot Tables. Demand and revenue rankings were calculated using sale filters, and return-risk rankings were calculated using cancellation filters.

Transaction_Type	Sale	Transaction_Type	Sale
Row Labels	**Product Revenue**	**Row Labels**	**Quantity sold**
DOT - DOTCOM POSTAGE	£206,248.77	23843 - PAPER CRAFT	80995
22423 - REGENCY CAKESTAND 3 TIER	£174,484.74	23166 - MEDIUM CER.	78033
23843 - PAPER CRAFT , LITTLE BIRDIE	£168,469.60	84077 - WORLD WAR	55047
85123A - WHITE HANGING HEART T-LIGHT HOLDER	£106,292.77	85099B - JUMBO BAG	48474
47566 - PARTY BUNTING	£99,504.33	85123A - WHITE HANC	37891
85099B - JUMBO BAG RED RETROSPOT	£94,340.05	22197 - POPCORN HO	36761
23166 - MEDIUM CERAMIC TOP STORAGE JAR	£81,700.92	84879 - ASSORTED CO	36461
M - Manual	£78,112.82	21212 - PACK OF 72 R	36419
POST - POSTAGE	£78,101.88	23084 - RABBIT NIGHT	30788
23084 - RABBIT NIGHT LIGHT	£66,964.99	22492 - MINI PAINT SI	26633
Grand Total	**£1,154,220.87**	**Grand Total**	**467502**



DOTCOM POSTAGE generated the highest revenue (£206,248.77) from the lowest quantity of units (706 units), indicating a high-revenue/low-quantity profile including non-merchandise codes. REGENCY CAKESTAND 3 TIER was runner-up at £174,156.54 PAPER CRAFT, LITTLE BIRDIE was the number one by volume with 80,995 units and also made £168,469.60 but this was due to one very large transaction and should be treated with caution. WORLD WAR 2 GLIDERS ASSTD DESIGNS shows high volume, low revenue: 54,951 units only £13,814.01

Products with highest cancellation quantity:

Transaction_Type	Cancellation		
Row Labels	**Returned Quantity**	**Cancellation Records**	**Cancellation Value**
23843 - PAPER CRAFT , LITTLE BIRDIE	80995	1	£168,469.60
23166 - MEDIUM CERAMIC TOP STORAGE JAR	74494	10	£77,479.64
84347 - ROTATING SILVER ANGELS T-LIGHT HLDR	9376	4	£321.60
M - Manual	4066	244	£146,784.46
21108 - FAIRY CAKE FLANNEL ASSORTED COLOUR	3150	3	£6,591.42
20971 - PINK BLUE FELT CRAFT TRINKET BOX	2617	5	£2,778.77
85123A - WHITE HANGING HEART T-LIGHT HOLDER	2578	42	£6,624.30
21175 - GIN + TONIC DIET METAL SIGN	2030	7	£3,775.33
22920 - HERB MARKER BASIL	1527	2	£841.05
22273 - FELTCRAFT DOLL MOLLY	1447	6	£3,512.65
Grand Total	**182280**	**324**	**£417,178.82**

Cancellation exposure is concentrated in a limited number of codes. PAPER CRAFT, LITTLE BIRDIE and MEDIUM CERAMIC TOP STORAGE JAR are max cancel quantities. The highest value is AMAZON FEE at £235,281.59 followed by PAPER CRAFT, LITTLE BIRDIE and Manual adjustments. In the evaluation of the management of operational product returns, these service and adjustment codes should be separated from physical products.

Customer Analysis

The customer analysis was limited to sale transactions with a known CustomerID. The PivotTable calculated total revenue, number of distinct invoices, quantity purchased, earliest purchase date and latest purchase date. average order value was computed as customer revenue divided by unique invoices. Recency was calculated based on a fixed analysis date of 10 December 2011, one day after the latest transaction, so that the results are not subject to change

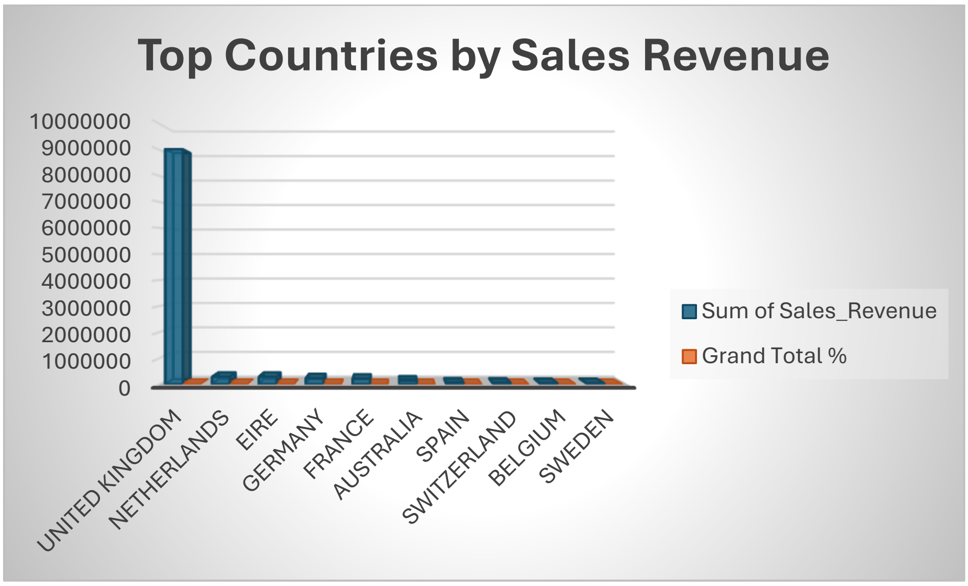
=INT(MAXIFS(table13[InvoiceDate],table13[Transaction_Type],"Sale"))+1

Customer 14646 was the highest revenue generator at £280,206.02, with customer 18102 in second place at £259,657.30. Customer 14911 has completed 201 sale invoices indicating a high frequency of repeat purchasing. Customer 16446 has a very high average order value of £84, 236.25 as only two invoices made up £168, 472.50. The recency values for the top customers are low which implies that most of them bought at the end of the observation period.

Country Analysis

Country performance was based on sale revenue only; Country in PivotTable Rows and Sales_Revenue in Values. The ranking was sorted in descending order and used to build the dashboard country chart.

Row Labels	Sum of Sales_Revenue	Grand Total %
United Kingdom	9025222.084	87.04%
Netherlands	285446.34	2.75%
EIRE	283453.96	2.73%
Germany	228867.14	2.21%
France	209715.11	2.02%
Australia	138521.31	1.34%
Spain	61577.11	0.59%
Switzerland	57089.9	0.55%
Belgium	41196.34	0.40%
Sweden	38378.33	0.37%
Grand Total	10369467.62	100.00%



The United Kingdom’s contribution was £9,001,744.09 which was 84.59% of total sales revenue. The Netherlands and EIRE were the next biggest markets but each accounted for less than 3% of the total. The concentration implies that the overall performance is highly sensitive to the UK market

Scenario Testing:

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Scenario 1 results in net revenue of £9,260,724.52. The larger impact is from scenario 2 as the decrease in sale quantity by 10% decreases revenue whilst the cancellation value is unaffected resulting in a net revenue of £8,683,919.99. In all three cases December 2011 is the lowest net revenue month but this is a partial month and must therefore be interpreted with caution.

Dashboard Construction

The Dashboard sheet was created for a management audience that is non-technical. It shows five KPI cards, and below the charts for monthly sales, products, countries, average order value, and scenario performance. PivotCharts are connected to the right PivotTables, and you can use slicers to filter the dashboard by quarter and month. All Supporting PivotTables should be refreshed prior to submission to ensure the figures displayed match Clean_Data

Assumptions and Limitation:

All eight original fields were used to remove exact duplicates; the Raw_Data sheet was not altered.

Transactions with Missing CustomerID were kept for overall analysis, but not included in customer level calculations.

Descriptions missing were found and removed from the named-product rankings.

Rows with UnitPrice <= 0 were classified as Invalid Price and did not contribute to revenue or cancellation value.

A cancellation was detected when the invalid-price rule was used, i.e. InvoiceNo began with C or Quantity was negative.

The cancellation value was represented as a positive value in ABS, which allowed it to be easily deducted from the sales revenue.

Date was saved as a real date with the first of the month to allow proper chronological sort.

Scenario 2 decreased sale revenue by 10% and did not change historical cancellation value.

December 2011 contains only nine transaction days and should not be compared directly to full months.

Conclusion:

We removed 5,268 exact duplicate rows, resulting in 536,641 unique transactions to analyze. Sales revenue was £10,642,110.80. After cancellations of £893,979.73, the sales revenue was £9,748,131.07. Sales peaked in November 2011 and were focused in the United Kingdom. The product and customer results show that rankings can be materially affected by a small number of transactions, service codes, and high-value customers. The scenario model indicates a 10% reduction in sales quantities results in a larger decline in net revenue than a 5% decrease in price, given the assumptions documented. The checks and dashboard in the workbook provide a traceable foundation